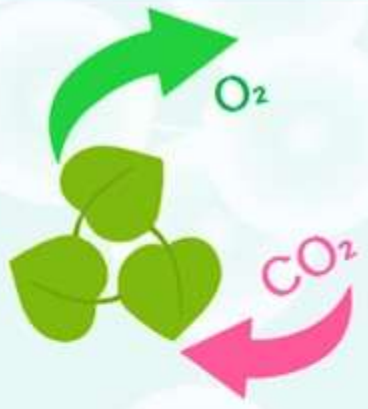


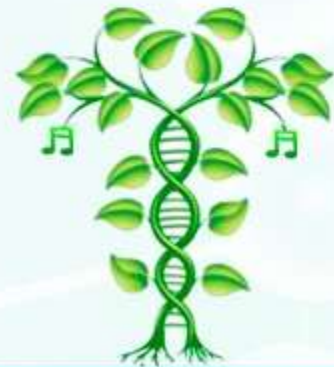


BIOLOGY



MICRONUTRIENTS

MODULE - 3





FOOD WE EAT

NUTRIENTS

DIETARY FIBRES

WATER

MACRO-NUTRIENTS



MICRO-NUTRIENTS



MICRONUTRIENTS

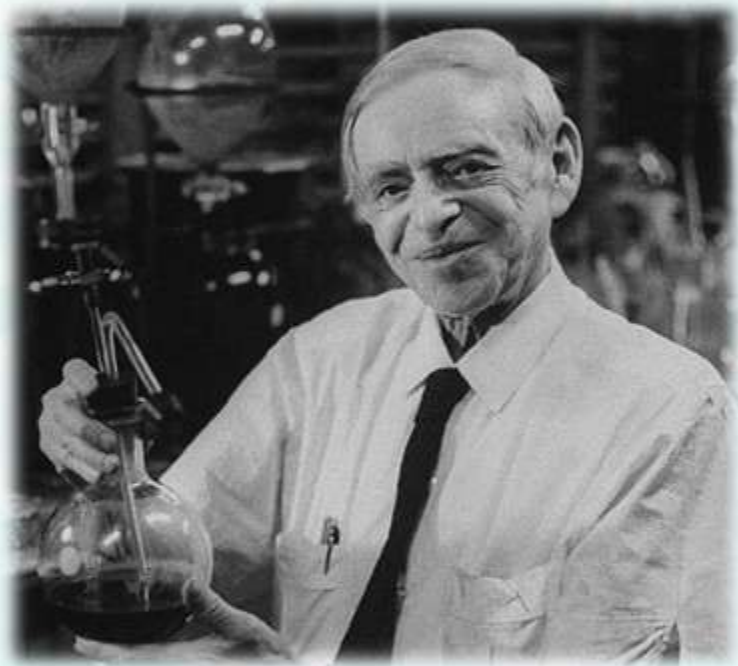
VITAMINS



ASSISTS IN OVERALL GROWTH AND FUNCTION



MICRONUTRIENTS



Casimir Funk

FACTS ABOUT VITAMINS

- 1 Vitamin was first named as *Vitamine* in 1912 (vital+amine) by Casimir Funk means vital for health.
- 2 Vitamins are organic compounds that our body can't make itself and needs in the diet.
- 3 Vitamins have **diverse biochemical functions** -essential for normal growth and development of a human body.
- 4 These are used as an overall maintenance tool for cells, tissues and organs.
- 5 Assists in the uptake and absorption of other nutrients like carbohydrate and proteins.



VITAMINS - TYPES

VITAMIN A

VITAMIN B

VITAMIN C

VITAMIN D

VITAMIN E

VITAMIN K

- Vitamin B and Vitamin C are water soluble.
- Vitamin A, Vitamin D, Vitamin K, Vitamin E are Fat soluble.



QUESTIONS ON VITAMINS

Q. The vitamins, which are soluble in water, are

(a) A and B

☒ (c) B and C

(b) C and D

(d) A and D

WATER SOLUBLE

VITAMIN B

VITAMIN C



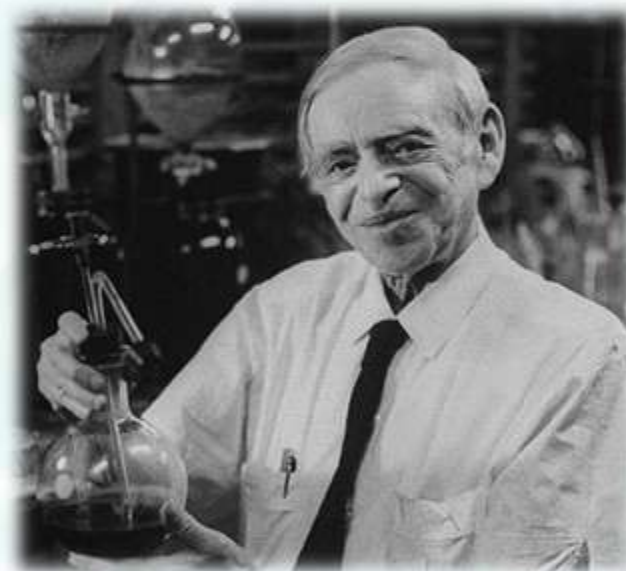
Q. Who coined the term Vitamine in 1912?

(a) Peter Debye

☒ (b) Casimir Funk

(c) James Watt

(d) Fritz Klatte



CASIMIR FUNK

MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN A



CHEMICAL NAME	SOLUBLILITY	FUNCTION	SOURCE
RETINOL, CAROTENOIDS including β -CAROTENE	FAT	Promotes growth and important component for retinal cells (rhodopsin)	▪ Pumpkins ▪ Spinach ▪ Carrots



QUESTIONS ON VITAMINS

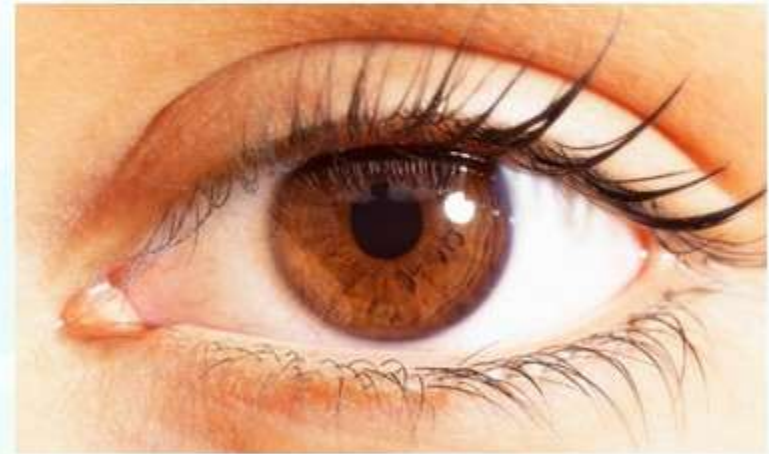
Q. Which vitamin is made up of the chemical component β -CAROTENE?

- ☒ (a) Vitamin A
- (c) Vitamin C
- (b) Vitamin B1
- (d) Vitamin B3



Q. What is the deficiency disease of Vitamin A called?

- (a) Beri- Beri
- (b) Pellagra
- (c) Scurvy
- ☒ (d) Night-blindness



VITAMINS - TYPES



VITAMIN B-complex

VITAMIN B

- 1 All B vitamins help to produce energy, to convert carbohydrates into glucose.
- 2 Also perform and promote various other functions in the body.
- 3 B vitamins are water soluble that means they will not be stored in body but flushed out with water .
- 4 That is why we need regular intake of vitamin B.
- 5 There are eight B vitamins known as Vitamin B complex.



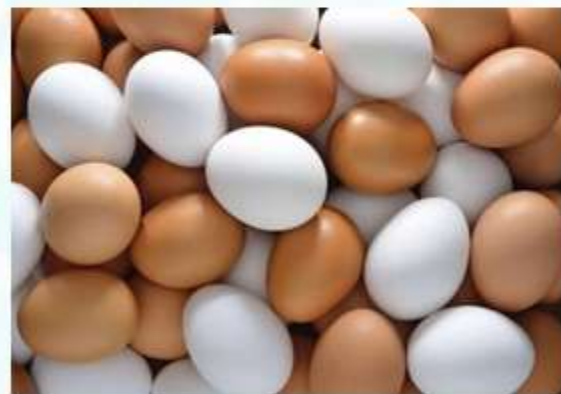
MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN B1



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
THIAMINE	WATER	Normal functioning of heart brain and nervous system	▪ Brown rice ▪ Whole grains ▪ Eggs



Helps in improving muscles !



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN B2



CHEMICAL NAME	SOLUBLILITY	FUNCTION	SOURCE
RIBOFLAVIN	WATER	Regulates breakdown of food and red blood cell production	▪ Liver, ▪ Green vegetables ▪ Eggs ▪ Yeast



Helps in improving muscles !



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN B3



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
NIACIN	WATER	Promotes health of skin and functioning nervous system and digestive system	- Lean meat - groundnut - Milk



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN B5



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
PANTOTHENIC ACID	WATER	Manufacture of RBC and hormones	- Lean meat - Brocoli - Mushroom



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN B6



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
PYRIDOXINE	WATER	Amino acid uptake and synthesis of neurotransmitters in brain	▪ Fish ▪ Liver ▪ Potatoes



Also called as the Brain Vitamin



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN B7



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
BIOTIN	WATER	Breakdown of carbohydrate, fats and proteins	▪ Egg yolks ▪ Almonds ▪ Nuts ▪ Whole grain



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN B9



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
FOLIC ACID	WATER	Production of RBC, development of brain and spinal cord(foetus)	▪ Liver ▪ Leafy vegetables like spinach



Also called as the Brain Vitamin



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN B12



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
COBALAMINE	WATER	Formation of RBC and functioning of CNS via the synthesis of myelin	▪ Mackerels ▪ Sardines ▪ Dairy products ▪ Coconut



Also called as the Brain Vitamin



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN C



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
ASCORBIC ACID	WATER	Forms an important protein to form skin, ligaments & blood vessels	▪ Citrus fruits ▪ Liver ▪ Capsicum



Readily gets destroyed by heat



MCQs ON VITAMINS

Which of the following statements are correct regarding Vitamin- C?

- 1) Its deficiency causes Swelling of Gums.
- 2) It gets easily destroyed by heat during cooking.
- 3) Used in treatment of chronic diseases.
- 4) Large doses may cause gastro intestinal discomfort and headache.

- a) 1, 2 & 3 only
c) 1 & 2 only

- b) 2 & 3 only
d) 1, 2, 3 & 4



MCQs ON VITAMINS

Q. What is the chemical form of Vitamin C?

- (a) Carotene (b) Riboflavin
☒ (c) Ascorbic acid (d) Pantothenic acid



Q. Which Vitamin is present in brown rice?

- (a) Vitamin A ☒ (b) Vitamin B
(c) Vitamin C (d) Vitamin D



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN D



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
ERGOCALCIFEROL(D2) CHOLECALCIFEROL(D3)	FAT	Helps the body to use calcium and phosphorus	▪ Fish eggs ▪ Mushrooms. ▪ Produced by UV rays in the skin



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN E



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
TOCOPHEROLS	FAT	Prevents cellular oxidation thus reduces ageing	- Almonds - Sunflower seeds - Sweet potato



MICRONUTRIENTS

VITAMINS - TYPES

VITAMIN K



CHEMICAL NAME	SOLUBILITY	FUNCTION	SOURCE
PHYLLOQUINONE	FAT	Maintaining normal clotting of blood	▪ Cabbage ▪ Spinach ▪ Egg yolks



MICRONUTRIENTS

MINERALS



 Learning Space
Maintains overall health of an organism



MINERALS

MACRONUTRIENT MINERALS

REQUIRED BY OUR BODY IN
LARGE AMOUNTS

Potassium

Magnesium

Calcium

Sodium

Chlorine

Phosphorus

Sulphur

MICRONUTRIENT MINERALS

REQUIRED BY OUR BODY IN
SMALL OR TRACE AMOUNTS

Iron

Iodine

Cobalt

Copper

Zinc

Manganese

Fluorine

Molybdenum



MINERALS

MACRONUTRIENT MINERALS IN OUR FOODS



Potassium



Calcium



Chlorine



Sulphur



Magnesium



Sodium



Phosphorus



MINERALS

MICRONUTRIENT MINERALS IN OUR FOODS



Iron



Cobalt



Copper



Zinc



Molybdenum



Fluorine



Manganese



Iodine





FOOD WE EAT

NUTRIENTS

✓ DIETARY FIBRES

✓ WATER

MACRO -
NUTRIENTS



MICRO -
NUTRIENTS



DIETARY FIBRE/ ROUGHAGE



INDIGESTABLE PART OF OUR FOOD BUT HELPS IN CLEANING
BOWELS AND MAINTAINING A PROPER DIGESTIVE HEALTH



DIETARY FIBRE/ ROUGHAGE

SOLUBLE FIBRE

- ✓ Dissolves in water to form a gel like material
- ✓ Helps reduce blood cholesterol
- ✓ Helps reduce blood glucose levels



INSOLUBLE FIBRE

- ✓ Promotes the movement of food through the digestive system
- ✓ Increases stool bulk
- ✓ Reduces cramps and constipation



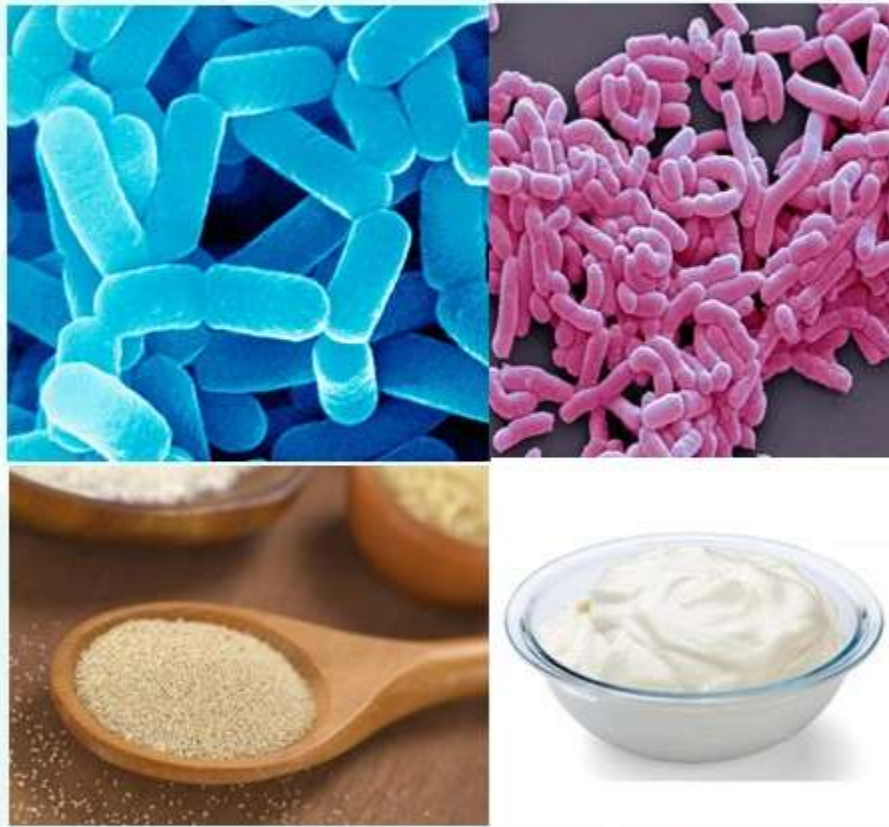
WATER: A GREAT SOURCE OF MINERAL !!!



- 1 Our body is made up of 70% water thus it is the most important component of food.
- 2 Water is a storehouse of essential minerals like calcium, potassium, iron, magnesium etc.
- 3 The entire digestive system is dependent on water as all the enzymes that digest food can only act on water surface.
- 4 Water can also dissolve many harmful impurity
- 5 Our diet should not be complete without 8 to 10 glasses of water everyday (min. 2liter)



HOW MICROORGANISMS SERVE AS A NUTRIENT?



PROBIOTICS

- 1 Probiotics are live bacteria and yeast that is good for our health and digestive system.
- 2 Probiotics help to move food properly through the digestive tracts and smooth the intestinal irritation.
- 3 Important sources of probiotics are curd, cheese dark chocolate etc.



HOW MUCH NUTRIENTS SHALL WE INTAKE?



- *For growth and maintenance of good health our diet should contain all the nutrients that our body needs in right quantities.*
- *It also should contain good amount of roughage and water.*
- *Such a diet is called **Balanced diet***



HOW MUCH NUTRIENTS SHALL WE TAKE?

THE 3 ASPECTS OF DIET

UNDERNUTRITION

- Not enough nutrition goes to the body and reduced proportion of carbohydrate proteins or vitamins.



BALANCED NUTRITION

- Right balance of all the nutrition goes to the body.
- The proportion of all the nutrients are measured according to the body requirements



OVER NUTRITION

- Too much nutrition goes to the body .
- Increased proportion of carbohydrate proteins or vitamins is taken.



Malnutrition is a broad term which refers to both under nutrition or over nutrition

BALANCED DIET IN HUMANS CHANGES WITH THE CHANGE IN AGE GENDER AND WORKING CONDITIONS

YOUNG AGE



- 1 Proteins- milk and milk products
- 2 Carbohydrates: starchy carbohydrate with less fiber
- 3 Vitamins and minerals: fruits and vegetables
- 4 Healthy fats in moderate amounts

MIDDLE AGE



- 1 Carbohydrates like cereals potato
- 2 Protein from meat and milk products
- 3 Vitamins and minerals: fruits and green leafy vegetables
- 4 Healthy fats such as nuts

OLD AGE



- 1 Proteins : milk and meat products
- 2 Vitamins and minerals: fruits and vegetables & calcium
- 3 Carbohydrates from brown rice and whole wheat
- 4 Healthy fats in moderate amounts



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

Amrita Naidu
D. Sunil Kumar
L. V. Krishna
D. Chaitanya

K. Srikanth
G. Ravi Babu
D. Joji
K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2
0 9 8 4 9 9 4 2 2 9 9